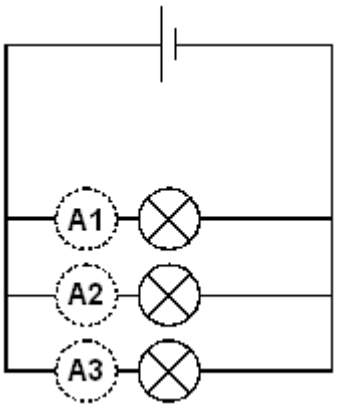
Q1. Peter measured the current through each of three similar bulbs in a parallel circuit.



He had only one ammeter and he placed it first at A1, then A2, then A3, in order to measure the currents.

The table shows his results.

position of ammeter	current, in amps
A1	0.14
A2	0.16
A3	0.15

(a) He expected the current readings to be the **same** for each bulb but found they were **different**.

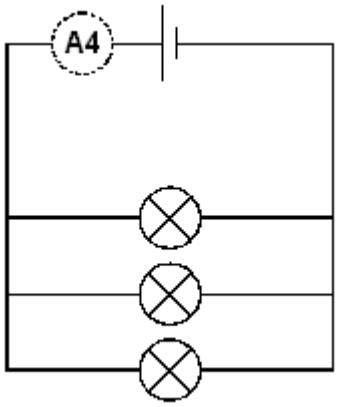
Suggest **two** reasons why the readings were different.

1. .....

2. .....

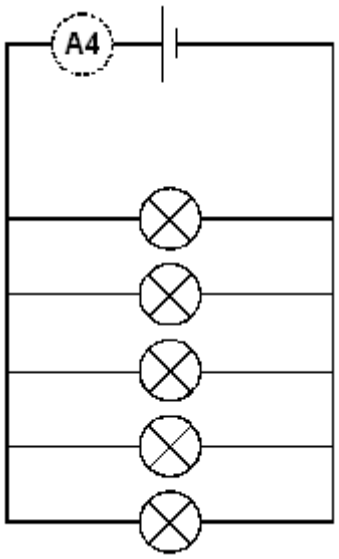
2 marks

(b) Peter then measured the current at **A4** and recorded it as 0.45 A. He concluded that the current at **A4** could be calculated by adding together the currents through each of the bulbs at positions **A1**, **A2** and **A3.**



He added two more similar bulbs to his circuit, in parallel. The current through each bulb was 0.15 A.
Use Peter's conclusion to predict the current at **A4** with the 5 bulbs in the circuit.

..... A



1 mark

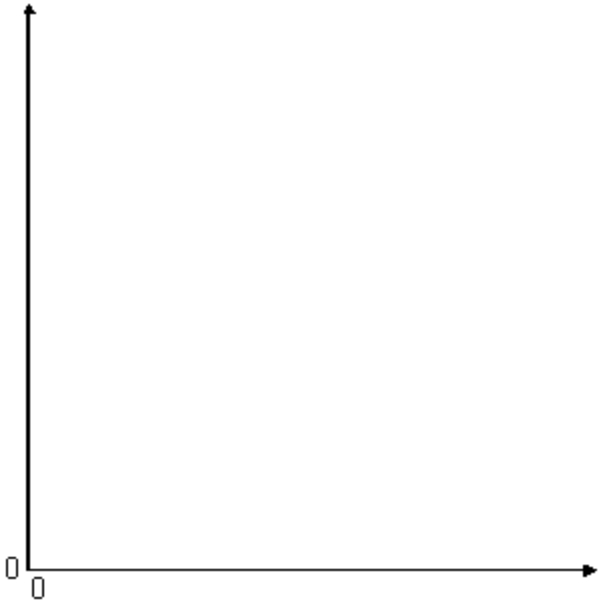
(c) Peter left the circuit connected overnight. He used a datalogger to measure the current at position **A4** at regular intervals of time. The next morning the bulbs were dim.

Using the axes below, sketch (do **not** plot) how the current at position **A4** might change with time.

Indicate on the graph:

(i) The correct labels for each axis, including the correct units.

(ii) The shape of the graph you would expect to obtain.



2 marks
Maximum 5 marks

Mark schemes

Q1.

(a) any **two** from:

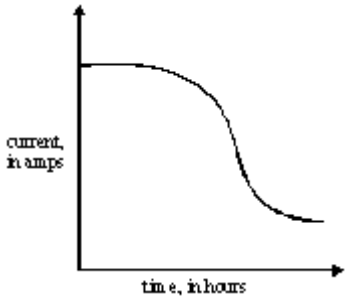
- manufacturing differences **or** bulbs are different
  accept 'different resistances'
  accept 'different ages'
- reading error
- dirty contacts
  accept 'bulbs were not screwed in properly'
- unreliable **or** inaccurate meter
  accept 'faulty ammeter'
  *accept 'different wires' **or** 'differences in the wires'*

2 (L6)

(b) 0.75

1 (L6)

(c)



***both** axes must be labelled correctly with both the variable and the unit*

Y axis : current, in amps **or** A **or** milliamps **or** mA
accept 'I , in amps'

X axis : time, in hours **or** minutes **or** seconds
accept 't, in hours'

1 (L7)

a line **or** curve from top left to bottom right

1 (L7)

[5]